

Supplementary material

The impacts of air pollution on maternal stress during pregnancy

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Supplementary Table 1. Spearman correlation coefficients between air pollutants and weather variables in this study

	Relative humidity	Wind speed	SO ₂	NO ₂	PM ₁₀
Temperature	-0.02	0.27**	0.17**	-0.01	0.03
Relative humidity		-0.06**	-0.34**	-0.06*	-0.34**
Wind speed			-0.28**	-0.44**	-0.33**
SO ₂				0.60**	0.68**
NO ₂					0.65**

** $P < 0.01$

* $P < 0.05$

Supplementary Table 2. Increased risk [OR (95%CI)] of high maternal emotional stress (GSI: P75-P100) for an IQR increase in SO₂, NO₂, and PM₁₀, and the modifying effect of complications of pregnancy^a

Lag days	SO ₂	NO ₂	PM ₁₀
Lag 0			
Total	1.30 (1.11, 1.52)*	1.09 (0.91, 1.29)	1.16 (1.02, 1.34)*
With complications of pregnancy	1.52 (0.94, 2.44)	0.99 (0.58, 1.69)	1.21 (0.78, 1.86)
Without complications of pregnancy	1.28 (1.08, 1.52)*	1.10 (0.92, 1.33)	1.16 (1.00, 1.34)
Lag 0-2			
Total	1.15 (0.95, 1.40)	1.11 (0.91, 1.35)	1.09 (0.96, 1.23)
With complications of pregnancy	1.25 (0.69, 2.28)	1.08 (0.59, 1.98)	1.06 (0.70, 1.59)
Without complications of pregnancy	1.14 (0.93, 1.39)	1.11 (0.90, 1.36)	1.09 (0.96, 1.25)
Lag 0-5			
Total	1.21 (0.99, 1.49)	1.34 (1.05, 1.70)*	1.22 (1.00, 1.48)
With complications of pregnancy	1.08 (0.57, 2.08)	1.24 (0.60, 2.53)	1.19 (0.63, 2.24)
Without complications of pregnancy	1.23 (0.99, 1.53)	1.35 (1.04, 1.74)*	1.23 (1.00, 1.51)
Lag 0-7			
Total	1.17 (0.92, 1.48)	1.26 (0.96, 1.66)	1.16 (0.91, 1.48)
With complications of pregnancy	0.87 (0.42, 1.84)	1.16 (0.51, 2.62)	0.89 (0.41, 1.94)
Without complications of pregnancy	1.23 (0.95, 1.57)	1.27 (0.95, 1.70)	1.22 (0.94, 1.59)
Lag 0-14			
Total	1.23 (0.94, 1.61)	1.23 (0.90, 1.69)	0.99 (0.70, 1.38)
With complications of pregnancy	1.17 (0.50, 2.73)	1.43 (0.54, 3.82)	0.78 (0.25, 2.40)
Without complications of pregnancy	1.27 (0.95, 1.69)	1.20 (0.86, 1.68)	1.07 (0.75, 1.53)

* $P < 0.05$

^a Adjusted for maternal age, education, occupation, complications of pregnancy, abortion history, family monthly income, type of family structure, average temperature, relative humidity, and wind speed.